

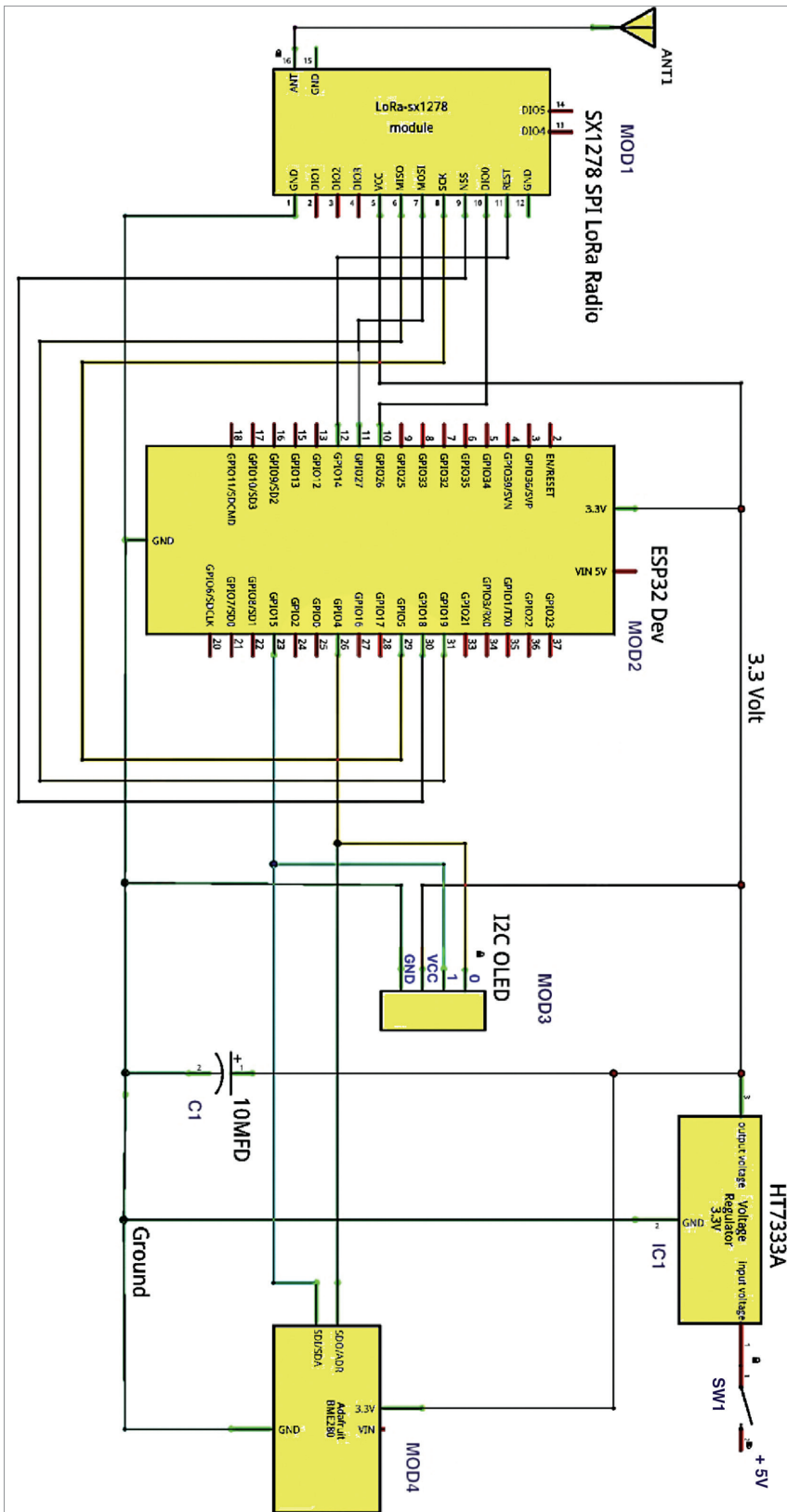


Fig. 3: Circuit diagram

flasher.Meshtastic.org/. This link works only in Chrome or Edge browser.

The circuit diagram of the LoRa project is shown in Fig. 3. It is built around SX1278 SPI LoRa radio (MOD1), ESP32 development board (MOD2), I2C OLED (MOD3), BME280 (MOD4), 3.3V HT7333A regulator (IC1), and a few other components.

## Flashing firmware

In the first screen, select the firmware version you want to flush with (Heltec V1). In the next screen, select the CP2102 USB to UART bridge, which is the interface hardware that sits in between the MCU and the computer. Next, follow the instructions that appear on the screen—like warning for erasing the entire device, etc.

Ensure the internet connection is stable during this crucial 4 to 5 minutes duration. For the built-in Heltec board you may need to keep the 'Boot' button pressed for the entire 4 to 5 minutes of the uploading phase.

## Uploading software

By command-line-instruction method, you find that the browser-based firmware updation does not work reliably. Many times, it shows that it has completed the task very quickly, but it has not. It takes about 4 to 5 minutes time for installing/updating the firmware. That is why both the firmware and the command-line-instruction methods are provided here.

### Mac commands/Linux commands

```
pip3 install --upgrade esptool [upgrade esptool first]
esptool.py chip_id [you will get a long output with chip id and chip mac id]
cd ~/Downloads/firmware/ [firmware-heltec-v1-2.2.12.092e6f2.bin]
./device-install.sh -f firmware-BOARD-VERSION.bin [for installing]
./device-update.sh -f firmware-BOARD-VERSION.bin [for updating]
```

### Windows commands

```
device-install.bat -f firmware-BOARD-
```